

A Period-Three Limit Orbit in a Thue–Morse Heteroclinic Closure

Route-A structural certificate C154

Abstract

We join a period-three past to a Thue–Morse future in one two-sided symbolic configuration and close its full shift orbit. The closure has exactly the Thue–Morse component, the interface orbit, and the three periodic phases. Consequently its only primitive orbit has length three and its Artin–Mazur zeta is $(1 - z^3)^{-1}$. This periodic layer is generated by one heteroclinic closure rather than freely attached.

1 The frozen interface

Let t_n be binary digit-sum parity for $n \geq 0$, and let $y_j = (2, 3, 4)_{j \bmod 3}$. Define

$$x_j = \begin{cases} y_j, & j < 0, \\ t_j, & j \geq 0, \end{cases} \quad X = \overline{\{\sigma^n x : n \in \mathbb{Z}\}},$$

where $(\sigma x)_j = x_{j+1}$. Positive shifts expose the Thue–Morse tail, whereas negative shifts expose one of the three phases of y . Thus

$$X = X_{\text{TM}} \sqcup \text{Orb}(x) \sqcup \text{Orb}(y). \quad (1)$$

The pair 40 occurs exactly at the interface, so every point of $\text{Orb}(x)$ is isolated and wandering. The full orbit of x is dense in X , while the period-three orbit is a proper closed invariant set; hence X is not minimal.

2 Periodic orbit ledger

The Thue–Morse language has no periodic point, and the wandering interface orbit has none. Therefore

$$\# \text{Fix}(\sigma^n) = \begin{cases} 3, & 3 \mid n, \\ 0, & 3 \nmid n. \end{cases} \quad (2)$$

Möbius inversion leaves three exact-period points at $n = 3$, hence one primitive cycle. Moreover

$$\log \zeta_X(z) = \sum_{q \geq 1} \frac{3z^{3q}}{3q} = -\log(1 - z^3), \quad \zeta_X(z) = \frac{1}{1 - z^3}. \quad (3)$$

The exact ledgers are implementation sentinels, not theorem cutoffs. The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL). We claim no target divisor, arithmetic/local factor, target functional equation or counting law, root number, automorphy, natural operator lift, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.